CURRENT ELECTRICITY

FORMULA SHEET | JEE MAINS | Let's Play With Physics

1. CURRENT & DRIFT VELOCITY

$$I = \frac{dq}{dt}, \qquad I = neAv_d$$

$$v_d = \frac{eE\tau}{m}, \qquad \mu = \frac{v_d}{E} = \frac{e\tau}{m}$$

Current density: $J = \dfrac{I}{A} = nev_d = \sigma E$

Vector form: $\vec{J} = \sigma\vec{E}, \quad I = \displaystyle\int \vec{J}\cdot d\vec{A}$

Typical values: $v_d \sim 10^{-4}$ m/s, $\tau \sim 10^{-14}$ s, free electrons in copper $n \approx 8.5 \times 10^{28}$ m^{-3}

⋆ Common Pitfall

- v_d is tiny, but the **signal** (field) travels at nearly c — that is why a bulb lights instantly
- Current is a **scalar** (adds algebraically); current density $\vec{J}$ is a vector
- Same I through a wire of varying area: $v_d \propto 1/A$, $J \propto 1/A$

2. OHM'S LAW & RESISTANCE

$$V = IR, \qquad R = \frac{\rho l}{A}$$

$$\rho = \frac{m}{ne^2\tau}, \qquad \sigma = \frac{1}{\rho}$$

Temperature: $R_T = R_0(1 + \alpha\,\Delta T)$

Metals $\alpha > 0$; semiconductors $\alpha < 0$; manganin/constantan $\alpha \approx 0$ (standard resistors)

Stretching a wire (volume constant):

- Length $\times n \Rightarrow R \to n^2 R \quad (R \propto l^2)$
- Radius $\div n \Rightarrow R \to n^4 R \quad (R \propto 1/r^4)$
- Small stretch: $\dfrac{\Delta R}{R} = 2\,\dfrac{\Delta l}{l}$

Colour	Digit	Colour	Digit
Black	0	Green	5
Brown	1	Blue	6
Red	2	Violet	7
Orange	3	Grey	8
Yellow	4	White	9
Tolerance: Gold ±5%, Silver ±10%, none ±20%			

Value = (1st 2nd) $\times 10^{3\text{rd}}$ Ω. Mnemonic: *B B ROY of Great Britain has a Very Good Wife*

⋆ Common Pitfall

- ρ depends on material & temperature only — **not** on length or area
- When a wire is stretched, use volume $= lA$ = constant, not A = constant

3. COMBINATION OF RESISTORS

Series: $R = R_1 + R_2 + \ldots$ (same I)

Parallel: $\dfrac{1}{R} = \dfrac{1}{R_1} + \dfrac{1}{R_2} + \ldots$ (same V)

Two in parallel: $R = \dfrac{R_1 R_2}{R_1 + R_2}$ n equal: nR or $\dfrac{R}{n}$

Current divider: $I_1 = I\,\dfrac{R_2}{R_1 + R_2}$

Network (each edge R)	R_{eq}
Cube, body diagonal	$5R/6$
Cube, face diagonal	$3R/4$
Cube, along an edge	$7R/12$
Infinite ladder (R series, R shunt)	$\frac{1+\sqrt{5}}{2}R$

⋆ Common Pitfall

- A wire with no resistance across a resistor **shorts** it — redraw first
- Points at the same potential (symmetry) can be joined or separated freely

4. EMF, INTERNAL RESISTANCE & CELLS

$$I = \frac{E}{R + r}$$

Discharging: $V = E - Ir$ Charging: $V = E + Ir$

Open circuit: $V = E$ Short circuit: $I = E/r$

$$r = R\left(\frac{E}{V} - 1\right)$$

Grouping	Current
n in series	$I = \dfrac{nE}{R + nr}$
m in parallel	$I = \dfrac{E}{R + r/m}$
Mixed (n series × m rows)	$I = \dfrac{nE}{R + nr/m}$

Mixed grouping gives **maximum current** when $R = \dfrac{nr}{m}$.

Two unequal cells in parallel:

$$E_{\text{eq}} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}, \qquad r_{\text{eq}} = \frac{r_1 r_2}{r_1 + r_2}$$

⋆ Common Pitfall

- One cell reversed in a series of n: $E_{\text{net}} = (n-2)E$, but $r_{\text{net}} = nr$
- While **charging**, terminal voltage is *more* than the emf

5. KIRCHHOFF'S LAWS

Junction law (KCL): $\sum I_{\text{in}} = \sum I_{\text{out}}$ — conservation of **charge**

Loop law (KVL): $\sum \Delta V = 0$ around any closed loop — conservation of **energy**

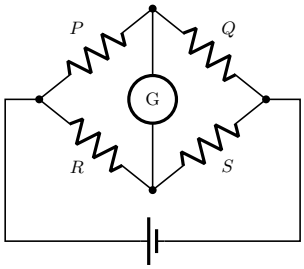
Sign rules (moving around the loop):

- Across R **along** the current: $-IR$; against it: $+IR$

- Across a cell from $-$ to $+$: $+E$; from $+$ to $-$: $-E$

A negative answer for I just means the real direction is opposite to the assumed one.

6. WHEATSTONE BRIDGE & METER BRIDGE



Balanced ($I_G = 0$): $\dfrac{P}{Q} = \dfrac{R}{S}$

Meter bridge: $\dfrac{R}{S} = \dfrac{l}{100 - l}$ (l in cm from the end beside R)

⋆ Common Pitfall

- In a balanced bridge, the galvanometer arm carries **no current** — remove it to find R_{eq}
- Interchanging cell and galvanometer keeps the bridge balanced
- Meter bridge is most accurate near the **middle** ($l \approx 50$ cm)

7. POTENTIOMETER

Potential gradient: $k = \dfrac{V_{\text{wire}}}{L}$, $\quad V = k\,l$

Compare emfs: $\dfrac{E_1}{E_2} = \dfrac{l_1}{l_2}$

Internal resistance: $r = R\left(\dfrac{l_1}{l_2} - 1\right)$

(l_1: cell alone, l_2: with R across the cell)

At balance the potentiometer draws **no current** $\Rightarrow$ it measures the true emf (an ideal voltmeter).

⋆ Common Pitfall

- No balance point if the driver cell emf is *less* than the emf being measured
- Longer wire (smaller k) $\Rightarrow$ more sensitive

8. ELECTRIC POWER & HEATING

$$P = VI = I^2R = \frac{V^2}{R}, \qquad H = I^2Rt$$

Rated bulb (P_0, V_0): $R = \dfrac{V_0^2}{P_0}$; on supply V:

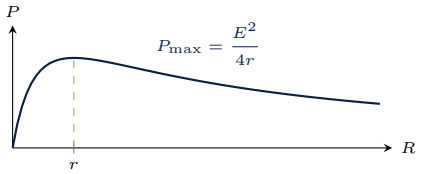
$P = P_0\left(\dfrac{V}{V_0}\right)^2$

1 kWh $= 3.6 \times 10^6$ J (1 "unit")

	Series	Parallel
Common	I	V
Use	$P = I^2R$	$P = V^2/R$
Brighter bulb	lower wattage	higher wattage
Total power*	$\dfrac{1}{P} = \sum \dfrac{1}{P_i}$	$P = \sum P_i$

*bulbs of equal rated voltage, used at that voltage

Maximum power transfer:



$P = \dfrac{E^2R}{(R+r)^2}$ is maximum when $R = r$: $\quad P_{\max} = \dfrac{E^2}{4r}$, efficiency $= 50\%$

Efficiency in general: $\eta = \dfrac{R}{R+r}$

⋆ Common Pitfall

- In series, the **lower**-wattage bulb glows brighter (it has more R)
- Max power transfer means 50% efficiency — **not** maximum efficiency

9. GALVANOMETER CONVERSIONS

Ammeter: shunt S in parallel, $\quad S = \dfrac{I_g G}{I - I_g}$

Voltmeter: R in series, $\quad R = \dfrac{V}{I_g} - G$

- Ideal ammeter: $R = 0$, in **series**; ideal voltmeter: $R = \infty$, in **parallel**
- Range of ammeter $\times n$: $S = \dfrac{G}{n-1}$; voltmeter $\times n$: add $(n-1)\,R_V$

10. JEE SHORTCUTS & MASTER TIPS

1. $I = neAv_d$; $\quad v_d = eE\tau/m$; $\quad J = \sigma E$
2. Stretch n times: $R \to n^2R$; radius halved: $R \to 16R$
3. $R_T = R_0(1 + \alpha\Delta T)$; for a metal $\rho \propto T$ (roughly)
4. Cells: $I = nE/(R + nr/m)$; max I at $R = nr/m$
5. Balanced bridge: $P/Q = R/S$; meter bridge $R/S = l/(100 - l)$
6. Potentiometer: $E_1/E_2 = l_1/l_2$; $\quad r = R(l_1/l_2 - 1)$
7. Max power at $R = r$: $P_{\max} = E^2/4r$
8. Bulbs: $R = V_0^2/P_0$; series $\to$ low watt brighter

⋆ Master Confusion List

1. **emf vs terminal voltage:** equal only when $I = 0$
2. **ρ vs R:** ρ is a material property; R depends on shape
3. **Drift speed vs signal speed:** 10^{-4} m/s vs nearly c
4. **KCL vs KVL:** charge conservation vs energy conservation
5. **Series bulbs:** lower wattage glows *brighter*
6. **Max power** ($R = r$) is not max efficiency ($R \gg r$)
7. **Potentiometer vs voltmeter:** potentiometer draws zero current at balance

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